

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I Pages 2–5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6–13

60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I

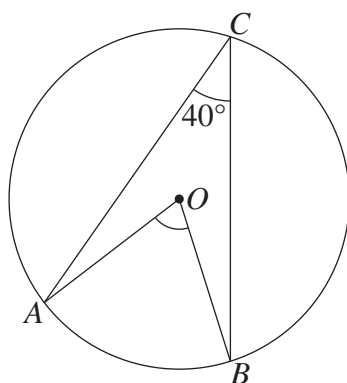
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 The points A , B and C lie on a circle with centre O , as shown in the diagram. The size of $\angle ACB$ is 40° .



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What is the size of $\angle AOB$?

- (A) 20°
 - (B) 40°
 - (C) 70°
 - (D) 80°
- 2 Which expression is equal to $\cos x - \sin x$?

- (A) $\sqrt{2} \cos\left(x + \frac{\pi}{4}\right)$
- (B) $\sqrt{2} \cos\left(x - \frac{\pi}{4}\right)$
- (C) $2 \cos\left(x + \frac{\pi}{4}\right)$
- (D) $2 \cos\left(x - \frac{\pi}{4}\right)$

3 What is the constant term in the binomial expansion of $\left(2x - \frac{5}{x^3}\right)^{12}$?

(A) $\binom{12}{3} 2^9 5^3$

(B) $\binom{12}{9} 2^3 5^9$

(C) $-\binom{12}{3} 2^9 5^3$

(D) $-\binom{12}{9} 2^3 5^9$

4 The acute angle between the lines $2x + 2y = 5$ and $y = 3x + 1$ is θ .

What is the value of $\tan \theta$?

(A) $\frac{1}{7}$

(B) $\frac{1}{2}$

(C) 1

(D) 2

5 Which group of three numbers could be the roots of the polynomial equation $x^3 + ax^2 - 41x + 42 = 0$?

(A) 2, 3, 7

(B) 1, -6, 7

(C) -1, -2, 21

(D) -1, -3, -14

6 What is the derivative of $3\sin^{-1}\frac{x}{2}$?

(A) $\frac{6}{\sqrt{4-x^2}}$

(B) $\frac{3}{\sqrt{4-x^2}}$

(C) $\frac{3}{2\sqrt{4-x^2}}$

(D) $\frac{3}{4\sqrt{4-x^2}}$

7 A particle is moving in simple harmonic motion with period 6 and amplitude 5.

Which is a possible expression for the velocity, v , of the particle?

(A) $v = \frac{5\pi}{3}\cos\left(\frac{\pi}{3}t\right)$

(B) $v = 5\cos\left(\frac{\pi}{3}t\right)$

(C) $v = \frac{5\pi}{6}\cos\left(\frac{\pi}{6}t\right)$

(D) $v = 5\cos\left(\frac{\pi}{6}t\right)$

- 8 In how many ways can 6 people from a group of 15 people be chosen and then arranged in a circle?

(A) $\frac{14!}{8!}$

(B) $\frac{14!}{8!6}$

(C) $\frac{15!}{9!}$

(D) $\frac{15!}{9!6}$

- 9 The remainder when the polynomial $P(x) = x^4 - 8x^3 - 7x^2 + 3$ is divided by $x^2 + x$ is $ax + 3$.

What is the value of a ?

(A) -14

(B) -11

(C) -2

(D) 5

- 10 Which equation describes the locus of points (x, y) which are equidistant from the distinct points $(a + b, b - a)$ and $(a - b, b + a)$?

(A) $bx + ay = 0$

(B) $bx + ay = 2ab$

(C) $bx - ay = 0$

(D) $bx - ay = 2ab$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Solve $\left(x + \frac{2}{x}\right)^2 - 6\left(x + \frac{2}{x}\right) + 9 = 0$. **3**

- (b) The probability that it rains on any particular day during the 30 days of November is 0.1. **2**

Write an expression for the probability that it rains on fewer than 3 days in November.

- (c) Sketch the graph $y = 6 \tan^{-1}x$, clearly indicating the range. **2**

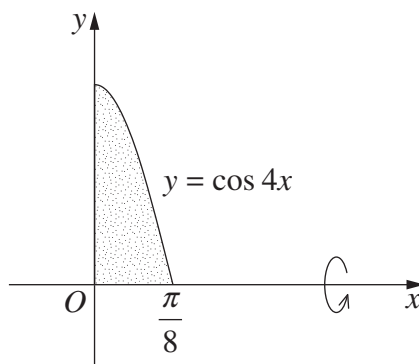
(d) Evaluate $\int_2^5 \frac{x}{\sqrt{x-1}} dx$ using the substitution $x = u^2 + 1$. **3**

(e) Solve $\frac{x^2 + 5}{x} > 6$. **3**

(f) Differentiate $\frac{e^x \ln x}{x}$. **2**

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) A particle is moving in simple harmonic motion about the origin, with displacement x metres. The displacement is given by $x = 2 \sin 3t$, where t is time in seconds. The motion starts when $t = 0$.
- (i) What is the total distance travelled by the particle when it first returns to the origin? **1**
- (ii) What is the acceleration of the particle when it is first at rest? **2**
- (b) The region bounded by $y = \cos 4x$ and the x -axis, between $x = 0$ and $x = \frac{\pi}{8}$, is rotated about the x -axis to form a solid. **3**



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Find the volume of the solid.

- (c) A particle moves along a straight line with displacement x m and velocity v m s⁻¹. The acceleration of the particle is given by **3**

$$\ddot{x} = 2 - e^{-\frac{x}{2}}.$$

Given that $v = 4$ when $x = 0$, express v^2 in terms of x .

Question 12 continues on page 8

Question 12 (continued)

- (d) Use the binomial theorem to show that

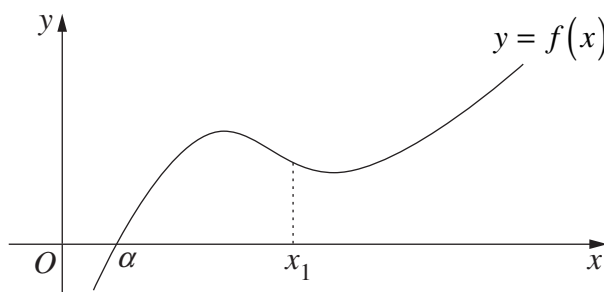
2

$$0 = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots + (-1)^n \binom{n}{n}.$$

- (e) The diagram shows the graph of a function $f(x)$.

1

The equation $f(x) = 0$ has a root at $x = \alpha$. The value x_1 , as shown in the diagram, is chosen as a first approximation of α .



A second approximation, x_2 , of α is obtained by applying Newton's method once, using x_1 as the first approximation.

Using a diagram, or otherwise, explain why x_1 is a closer approximation of α than x_2 .

- (f) Milk taken out of a refrigerator has a temperature of 2°C . It is placed in a room of constant temperature 23°C . After t minutes the temperature, $T^\circ\text{C}$, of the milk is given by

3

$$T = A - Be^{-0.03t},$$

where A and B are positive constants.

How long does it take for the milk to reach a temperature of 10°C ?

End of Question 12

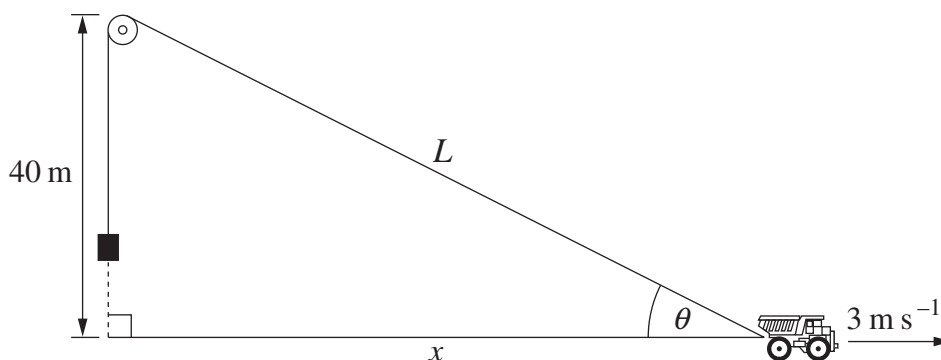
Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) Use mathematical induction to prove that $2^n + (-1)^{n+1}$ is divisible by 3 for all integers $n \geq 1$. **3**

- (b) One end of a rope is attached to a truck and the other end to a weight. The rope passes over a small wheel located at a vertical distance of 40 m above the point where the rope is attached to the truck.

The distance from the truck to the small wheel is L m, and the horizontal distance between them is x m. The rope makes an angle θ with the horizontal at the point where it is attached to the truck.

The truck moves to the right at a constant speed of 3 m s^{-1} , as shown in the diagram.



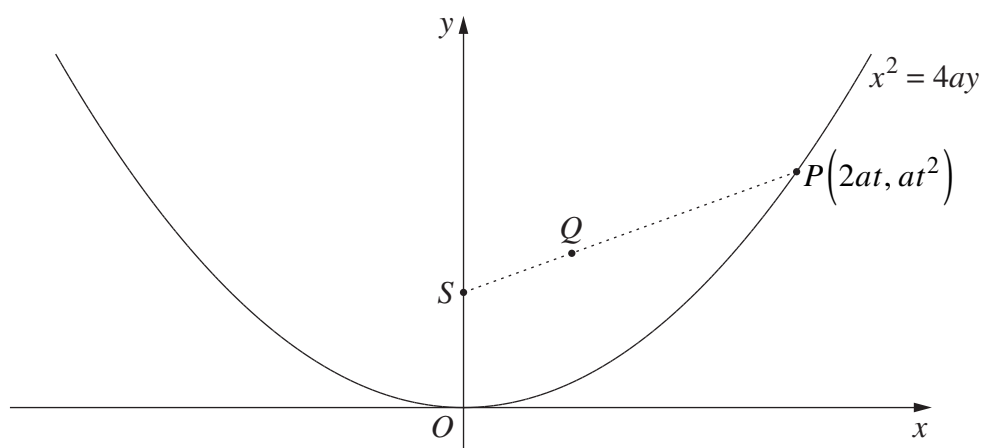
- (i) Using Pythagoras' Theorem, or otherwise, show that $\frac{dL}{dx} = \cos \theta$. **2**
- (ii) Show that $\frac{dL}{dt} = 3 \cos \theta$. **1**

Question 13 continues on page 10

Question 13 (continued)

- (c) The point $P(2at, at^2)$ lies on the parabola $x^2 = 4ay$ with focus S .

The point Q divides the interval PS internally in the ratio $t^2:1$.

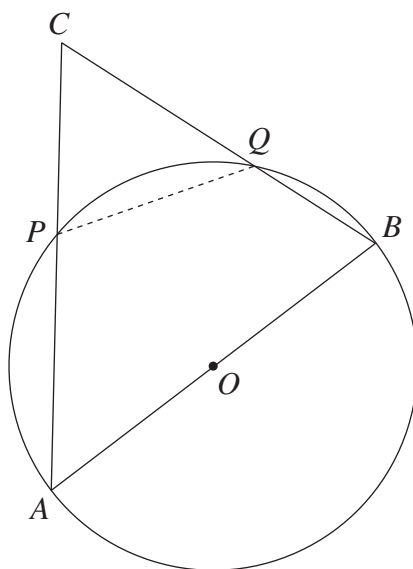


- | | | |
|-------|---|----------|
| (i) | Show that the coordinates of Q are $x = \frac{2at}{1+t^2}$ and $y = \frac{2at^2}{1+t^2}$. | 2 |
| (ii) | Express the slope of OQ in terms of t . | 1 |
| (iii) | Using the result from part (ii), or otherwise, show that Q lies on a fixed circle of radius a . | 3 |

Question 13 continues on page 11

Question 13 (continued)

- (d) In the diagram, AB is a diameter of a circle with centre O . The point C is chosen such that $\triangle ABC$ is acute-angled. The circle intersects AC and BC at P and Q respectively.



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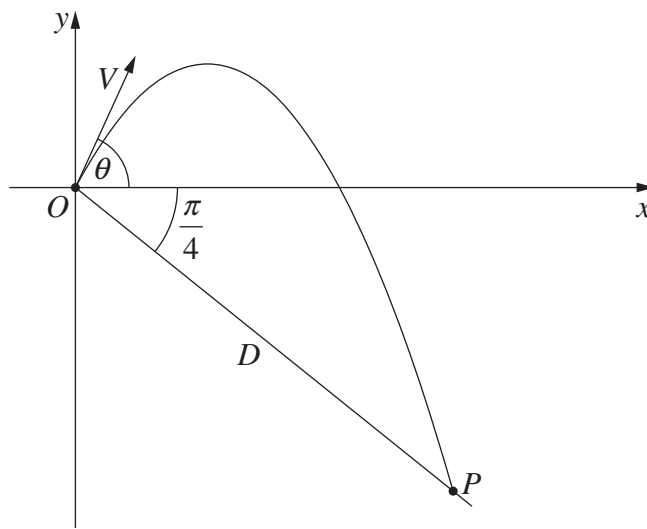
Copy or trace the diagram into your writing booklet.

- | | |
|---|----------|
| (i) Why is $\angle BAC = \angle CQP$? | 1 |
| (ii) Show that the line OP is a tangent to the circle through P , Q and C . | 2 |

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) The take-off point O on a ski jump is located at the top of a downslope. The angle between the downslope and the horizontal is $\frac{\pi}{4}$. A skier takes off from O with velocity $V \text{ m s}^{-1}$ at an angle θ to the horizontal, where $0 \leq \theta < \frac{\pi}{2}$. The skier lands on the downslope at some point P , a distance D metres from O .



The flight path of the skier is given by

$$x = Vt \cos \theta, \quad y = -\frac{1}{2}gt^2 + Vt \sin \theta, \quad (\text{Do NOT prove this.})$$

where t is the time in seconds after take-off.

- (i) Show that the cartesian equation of the flight path of the skier is given by 2

$$y = x \tan \theta - \frac{gx^2}{2V^2} \sec^2 \theta.$$

- (ii) Show that $D = 2\sqrt{2} \frac{V^2}{g} \cos \theta (\cos \theta + \sin \theta)$. 3

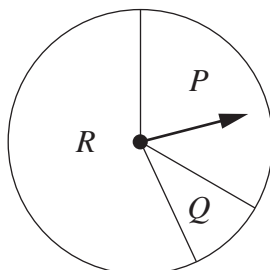
- (iii) Show that $\frac{dD}{d\theta} = 2\sqrt{2} \frac{V^2}{g} (\cos 2\theta - \sin 2\theta)$. 2

- (iv) Show that D has a maximum value and find the value of θ for which this occurs. 3

Question 14 continues on page 13

Question 14 (continued)

- (b) Two players A and B play a game that consists of taking turns until a winner is determined. Each turn consists of spinning the arrow on a spinner once. The spinner has three sectors P , Q and R . The probabilities that the arrow stops in sectors P , Q and R are p , q and r respectively.



The rules of the game are as follows:

- If the arrow stops in sector P , then the player having the turn wins.
- If the arrow stops in sector Q , then the player having the turn loses and the other player wins.
- If the arrow stops in sector R , then the other player takes a turn.

Player A takes the first turn.

- (i) Show that the probability of player A winning on the first or the second turn of the game is $(1 - r)(p + r)$. **2**

- (ii) Show that the probability that player A eventually wins the game is **3**

$$\frac{p + r}{1 + r}.$$

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

2014 HSC Mathematics Extension 1 Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	D
2	A
3	C
4	D
5	B
6	B
7	A
8	D
9	C
10	C

Section II

Question 11 (a)

Criteria	Marks
• Provides a correct solution	3
• Obtains a quadratic in x , or equivalent merit	2
• Attempts to solve quadratic in $x + \frac{2}{x}$, or equivalent merit	1

Sample answer:

$$\text{Let } n = x + \frac{2}{x}$$

$$n^2 - 6n + 9 = 0$$

$$(n - 3)^2 = 0 \quad \therefore n = 3$$

$$\text{Now, } x + \frac{2}{x} = 3$$

$$x^2 - 3x + 2 = 0$$

$$(x - 2)(x - 1) = 0$$

$$x = 2 \text{ or } x = 1$$

Question 11 (b)

Criteria	Marks
• Provides a correct solution	2
• Finds a correct expression for the probability of rain on 1 or 2 days, or equivalent merit	1

Sample answer:

$$P(\text{rain}) = 0.1 \quad P(\widetilde{\text{rain}}) = 1 - 0.1 = 0.9$$

Probability that it rains on fewer than 3 days is

$$\binom{30}{0}(0.9)^{30} + \binom{30}{1} \times 0.9^{29} \times 0.1 + \binom{30}{2} 0.9^{28} (0.1)^2$$

Question 11 (c)

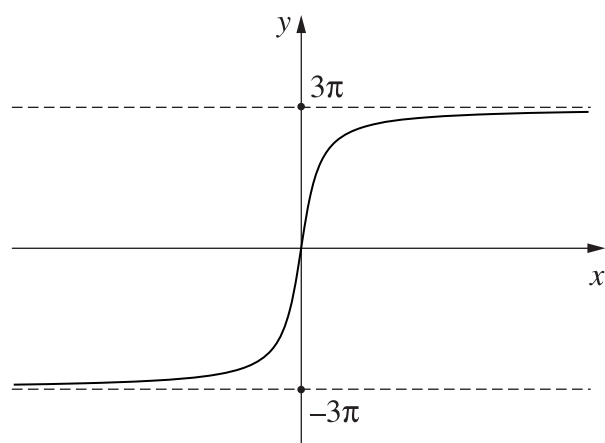
Criteria	Marks
• Draws a correct sketch	2
• Draws a graph showing the correct range or shape, or equivalent merit	1

Sample answer:

$$y = 6 \tan^{-1} x$$

$$\frac{-\pi}{2} < \frac{y}{6} < \frac{\pi}{2}$$

$$-3\pi < y < 3\pi$$



Question 11 (d)

Criteria	Marks
• Provides a correct solution	3
• Obtains a correct primitive, or equivalent merit	2
• Attempts the given substitution, or equivalent merit	1

Sample answer:

$$\int_2^5 \frac{x}{\sqrt{x-1}} dx$$

$$x = u^2 + 1 \quad | \quad x = 2 \quad u = 1$$

$$dx = 2u \, du \quad | \quad x = 5 \quad u = 2$$

$$\int_1^2 \frac{u^2 + 1}{u} 2u \, du$$

$$= 2 \left[\frac{u^3}{3} + u \right]_1^2$$

$$= 2 \left[\frac{8}{3} + 2 - \frac{1}{3} - 1 \right]$$

$$= 2 \left[\frac{10}{3} \right]$$

$$= \frac{20}{3}$$

Question 11 (e)

Criteria	Marks
• Provides a correct solution	3
• Obtains one correct interval, or equivalent merit	2
• Obtains a quadratic inequality, or equivalent merit	1

Sample answer:Multiply by x^2 :

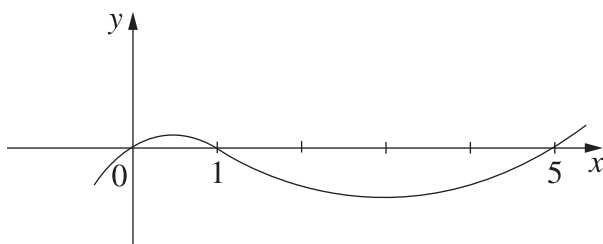
$$\frac{x^2(x^2 + 5)}{x} > 6 \times x^2$$

$$\text{ie } x(x^2 + 5) > 6x^2$$

$$x(x^2 + 5) - 6x^2 > 0$$

$$x(x^2 - 6x + 5) > 0$$

$$x(x - 5)(x - 1) > 0$$



$$\therefore 0 < x < 1 \quad \text{or} \quad x > 5$$

Question 11 (f)

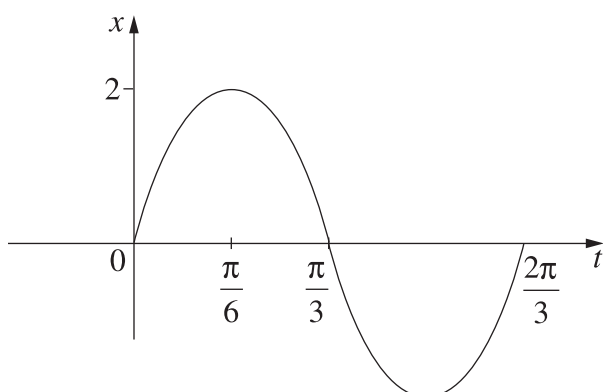
Criteria	Marks
• Finds the correct derivative	2
• Uses an appropriate differentiation rule, or equivalent merit	1

Sample answer:

$$y = \frac{e^x \ln x}{x}$$

$$\frac{dy}{dx} = \frac{x \left(e^x \cdot \frac{1}{x} + e^x \ln x \right) - e^x \ln x \cdot 1}{x^2}$$

$$= \frac{x \left(\frac{e^x}{x} + e^x \ln x \right) - e^x \ln x}{x^2}$$

Question 12 (a)**Question 12 (a) (i)**

Criteria	Marks
• Provides a correct answer	1

Sample answer:Distance travelled is $(2 + 2)$ m, ie 4 m.

Question 12 (a) (ii)

Criteria	Marks
• Provides a correct solution	2
• Identifies when the particle is first at rest, or equivalent merit	1

Sample answer:

When at rest,

$$\dot{x} = 0 \text{ ie } 6\cos 3t = 0$$

$$\therefore \cos 3t = 0$$

$$3t = \frac{\pi}{2}$$

$$t = \frac{\pi}{6}$$

$$\therefore \text{First at rest when } t = \frac{\pi}{6}$$

$$\text{Now, } x = 2\sin 3t$$

$$\dot{x} = 6\cos 3t$$

$$\ddot{x} = -18\sin 3t$$

$$\text{when } t = \frac{\pi}{6}$$

$$\ddot{x} = -18\sin \frac{\pi}{2}$$

$$= -18 \times 1$$

$$\text{acceleration} = -18 \text{ m/s}^2$$

Question 12 (b)

Criteria	Marks
• Provides a correct solution	3
• Obtains correct integral expression for volume in terms of $\cos 8x$, or equivalent merit	2
• Obtains integral expression for volume of the form $k \int_0^{\frac{\pi}{8}} \cos^2 4x \, dx$, or equivalent merit	1

Sample answer:

$$y = \cos 4x$$

$$V = \int_0^{\frac{\pi}{8}} \pi y^2 \, dx$$

$$= \int_0^{\frac{\pi}{8}} \pi \cos^2 4x \, dx$$

$$= \pi \int_0^{\frac{\pi}{8}} \frac{1 + \cos 8x}{2} \, dx$$

$$= \frac{\pi}{2} \left[x + \frac{\sin 8x}{8} \right]_0^{\frac{\pi}{8}}$$

$$= \frac{\pi}{2} \left[\frac{\pi}{8} + 0 - 0 - 0 \right]$$

$$\text{volume} = \frac{\pi^2}{16} \text{ unit}^3$$

Question 12 (c)

Criteria	Marks
• Provides a correct solution	3
• Obtains expression for v^2 possibly involving an undetermined constant	2
• Attempts to use $a = \frac{1}{2} \frac{d(v^2)}{dx}$ or equivalent merit	1

Sample answer:

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \ddot{x} = 2 - e^{\frac{-x}{2}}$$

$$\begin{aligned} \frac{1}{2} v^2 &= \int 2 - e^{\frac{-x}{2}} dx \\ &= 2x - \frac{e^{\frac{-x}{2}}}{-\frac{1}{2}} + c \\ &= 2x + 2e^{\frac{-x}{2}} + c \end{aligned}$$

When $x = 0, v = 4$

$$\frac{1}{2}(16) = 0 + 2 + c$$

$$c = 6$$

$$\therefore \frac{1}{2} v^2 = 2x + 2e^{\frac{-x}{2}} + 6$$

$$v^2 = 4x + 4e^{\frac{-x}{2}} + 12$$

Question 12 (d)

Criteria	Marks
• Provides a correct solution	2
• Attempts to use the binomial theorem.	1

Sample answer:

Consider $(1+x)^n$. From the binomial theorem:

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$$

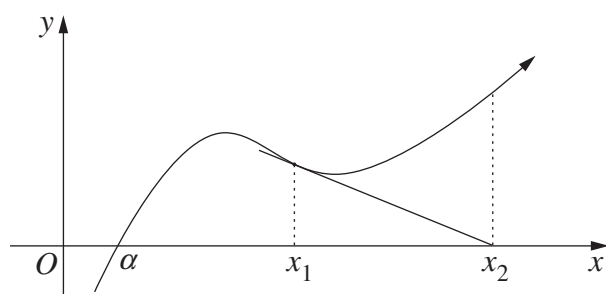
Let $x = -1$

$$(1+(-1))^n = 0 = \binom{n}{0} + \binom{n}{1}(-1) + \binom{n}{2}(-1)^2 + \dots + \binom{n}{n}(-1)^n$$

$$0 = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \dots + (-1)^n \binom{n}{n}$$

Question 12 (e)

Criteria	Marks
• Provides a valid explanation	1

Sample answer:

The tangent at $x = x_1$ is drawn, and it intersects the x axis at x_2 .

As shown in the diagram, x_2 is further away from α than x_1 .

Question 12 (f)

Criteria	Marks
• Provides a correct solution	3
• Obtains A and B , or equivalent merit	2
• Shows $A - B = 2$, or equivalent merit	1

Sample answer:

$$T = A - Be^{-0.03t}$$

$$\text{As } t \rightarrow \infty, \quad T \rightarrow 23$$

$$\therefore A = 23$$

$$\therefore T = 23 - Be^{-0.03t}$$

$$t = 0 \quad T = 2$$

$$2 = 23 - B$$

$$B = 21$$

$$\therefore T = 23 - 21e^{-0.03t}$$

Now, when $T = 0$

$$10 = 23 - 21e^{-0.03t}$$

$$21e^{-0.03t} = 13$$

$$e^{-0.03t} = \frac{13}{21}$$

$$-0.03t = \ln\left(\frac{13}{21}\right)$$

$$t = \frac{\ln\left(\frac{13}{21}\right)}{-0.03}$$

$$= 15.985... !$$

$$t \approx 16 \text{ minutes}$$

Question 13 (a)

Criteria	Marks
• Provides a correct proof	3
• Uses inductive assumption, or equivalent merit	2
• Establishes case for $n = 1$, or equivalent merit	1

Sample answer:

$$2^n + (-1)^{n+1}$$

$$n = 1, 2^1 + (-1)^2 = 2 + 1 = 3 \text{ which is divisible by 3}$$

Assume true for $n = k$ ie Assume $2^k + (-1)^{k+1}$ is divisible by 3ie $2^k + (-1)^{k+1} = 3M$ where M is an integer.We need to prove true for $n = k + 1$ ie That $2^{k+1} + (-1)^{k+2}$ is divisible by 3

$$\begin{aligned}
 2^{k+1} + (-1)^{k+2} &= 2(2^k) + (-1)^{k+2} \\
 &= 2[3M + (-1)^{k+2}] + (-1)^{k+2} \\
 &= 3[2M + (-1)^{k+2}]
 \end{aligned}
 \quad \left| \quad \begin{aligned}
 2^k &= 3M - (-1)^{k+1} \\
 &= 3M + (-1)^{k+2}
 \end{aligned}
 \right.$$

Which is a multiple of 3 since M and k are integers.ie if $2^k + (-1)^{k+1}$ is divisible by 3, then $2^{k+1} + (-1)^{k+2}$ is divisible by 3 $\therefore$ By the principle of mathematical induction $2^n + (-1)^{n+1}$ is divisible by 3 for all $n \geq 1$.

Question 13 (b) (i)

Criteria	Marks
• Provides a correct solution	2
• Uses Pythagoras' Theorem and attempts to differentiate, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 L &= \sqrt{40^2 + x^2} \\
 \frac{dL}{dx} &= \frac{1}{2} \cdot \frac{1}{\sqrt{40^2 + x^2}} \cdot 2x \\
 &= \frac{x}{\sqrt{40^2 + x^2}} \\
 &= \frac{x}{L} \\
 &= \cos \theta
 \end{aligned}$$

Question 13 (b) (ii)

Criteria	Marks
• Provides a correct solution	1

Sample answer:

$$\begin{aligned}
 \frac{dL}{dt} &= \frac{dL}{dx} \cdot \frac{dx}{dt} \\
 &= \frac{dL}{dx} \cdot 3 \\
 &= 3 \cos \theta \text{ (from part (i)).}
 \end{aligned}$$

Question 13 (c) (i)

Criteria	Marks
• Provides a correct solution	2
• Establishes one coordinate, or equivalent merit	1

Sample answer:

$$P(2at, at^2) \quad S(0, a)$$

$$t^2 : 1$$

$$Q = \left[\frac{0 \times t^2 + 2at}{t^2 + 1}, \frac{a \times t^2 + 1 \times at^2}{t^2 + 1} \right]$$

$$Q = \left[\frac{2at}{t^2 + 1}, \frac{2at^2}{t^2 + 1} \right]$$

Question 13 (c) (ii)

Criteria	Marks
• Provides a correct solution	1

Sample answer:

$$\begin{aligned}
 m_{OQ} &= \frac{\frac{2at^2}{t^2 + 1} - 0}{\frac{2at}{t^2 + 1} - 0} \\
 &= \frac{2at^2}{t^2 + 1} \times \frac{t^2 + 1}{2at} \\
 &= t
 \end{aligned}$$

Question 13 (c) (iii)

Criteria	Marks
• Provides a correct solution	3
• Obtains an equation in x , y and a	2
• Attempts to use part (ii), or equivalent merit	1

Sample answer:

$$x = \frac{2at}{t^2 + 1} \quad y = \frac{2at^2}{t^2 + 1}$$

$$m_{OQ} = t = \frac{y}{x} \text{ (from part (ii))}$$

$$\therefore x = \frac{2a \frac{y}{x}}{\frac{y^2}{x^2} + 1}$$

$$x = \frac{2axy}{y^2 + x^2}$$

$$x(x^2 + y^2) = 2axy, \text{ for } x \neq 0$$

$$x^2 + y^2 = 2ay$$

$$x^2 + y^2 - 2ay = 0$$

$$x^2 + y^2 - 2ay + a^2 = a^2$$

$$x^2 + (y - a)^2 = a^2$$

which is a circle of centre $(0, a)$ and radius a units and point Q lies on this circle.

Question 13 (d) (ii)

Criteria	Marks
• Provides a correct proof	2
• Observes that $\triangle OPA$ is isosceles, or equivalent merit	1

Sample answer:

From (i) $\angle OAP = \angle PQC = \alpha$

$\angle OAP = \angle OPA = \alpha$ ($OA = OP$ radii; angles opposite equal sides of $\triangle OAP$ are equal)

$\angle OPA = \angle TPC = \alpha$ (vertically opposite angles equal)

$\therefore \angle TPC = \angle PQC = \alpha$

$\therefore OP$ is a tangent to circle through P, Q, C , since the angle between a tangent (OP) and a chord (CP) is equal to the angle in the alternate segment.

Question 14 (a) (i)

Criteria	Marks
• Provides a correct solution	2
• Attempts to eliminate t , or equivalent merit	1

Sample answer:

$$x = Vt \cos \theta, y = -\frac{1}{2}gt^2 + Vt \sin \theta$$

$$t = \frac{x}{V \cos \theta}$$

$$y = -\frac{1}{2}g \cdot \frac{x^2}{V^2 \cos^2 \theta} + V \cdot \frac{x}{V \cos \theta} \cdot \sin \theta$$

$$= \frac{-gx^2}{2V^2} \sec^2 \theta + x \tan \theta$$

Question 14 (a) (ii)

Criteria	Marks
• Provides a correct solution	3
• Obtains the x coordinate of P , or equivalent merit	2
• Obtains an equation in x , or equivalent merit	1

Sample answer: P is the point of intersection of

$$y = x \tan \theta - \frac{gx^2}{2V^2} \sec^2 \theta \quad \text{and} \quad y = -x$$

$$\therefore -x = x \tan \theta - \frac{gx^2}{2V^2} \sec^2 \theta$$

$$x = \frac{gx^2}{2V^2} \sec^2 \theta - x \tan \theta$$

$$x(1 + \tan \theta) = \frac{gx^2}{2V^2} \sec^2 \theta$$

$$x \left[\frac{g \sec^2 \theta}{2V^2} - (1 + \tan \theta) \right] = 0$$

$$\therefore x = 0 \quad \text{or} \quad \frac{gx \sec^2 \theta}{2V^2} = 1 + \tan \theta$$

$$\frac{gx}{2V^2 \cos^2 \theta} = 1 + \frac{\sin \theta}{\cos \theta}$$

$$x = \frac{2V^2}{g} \cos^2 \theta + \frac{2V^2}{g} \frac{\sin \theta \cdot \cos^2 \theta}{\cos \theta} \quad (\cos \theta \neq 0)$$

$$= \frac{2V^2}{g} \cos \theta (\cos \theta + \sin \theta)$$

$$\text{Since } D^2 = 2x^2, \quad x = \frac{D}{\sqrt{2}}$$

$$\therefore \frac{D}{\sqrt{2}} = \frac{2V^2}{g} \cos \theta (\sin \theta + \cos \theta)$$

$$D = 2\sqrt{2} \frac{V^2}{g} \cos \theta (\sin \theta + \cos \theta)$$

Question 14 (a) (iii)

Criteria	Marks
• Provides a correct solution	2
• Uses double-angle formulae, or equivalent merit	1

Sample answer:

$$D = 2\sqrt{2} \frac{V^2}{g} [\cos^2 \theta + \sin \theta \cos \theta]$$

$$= 2\sqrt{2} \frac{V^2}{g} \left[\frac{1 + \cos 2\theta}{2} + \frac{1}{2} \sin 2\theta \right]$$

$$= \frac{2\sqrt{2}V^2}{2g} [1 + \cos 2\theta + \sin 2\theta]$$

$$\frac{dD}{d\theta} = \sqrt{2} \frac{V^2}{g} [-2\sin 2\theta + 2\cos 2\theta]$$

$$= 2\sqrt{2} \frac{V^2}{g} [\cos 2\theta - \sin 2\theta]$$

Question 14 (a) (iv)

Criteria	Marks
• Provides a correct solution	3
• Obtains $\theta = \frac{\pi}{8}$, or equivalent merit	2
• Attempts to solve $\frac{dD}{d\theta} = 0$, or equivalent merit	1

Sample answer:

D has a maximum value when $\frac{dD}{d\theta} = 0$

ie $\sin 2\theta = \cos 2\theta$

$$\tan 2\theta = 1$$

$$2\theta = \frac{\pi}{4}$$

$$\theta = \frac{\pi}{8}$$

$$\begin{aligned}\frac{d^2D}{d\theta^2} &= 2\sqrt{2} \frac{V^2}{g} (-2\sin 2\theta - 2\cos 2\theta) \\ &= -4\sqrt{2} \frac{V^2}{g} (\sin 2\theta + \cos 2\theta)\end{aligned}$$

$$\text{for } \theta = \frac{\pi}{8}, \sin 2\theta + \cos 2\theta > 0$$

$$\therefore \frac{d^2D}{d\theta^2} < 0$$

$$\therefore D \text{ is maximum when } \theta = \frac{\pi}{8}$$

Question 14 (b) (i)

Criteria	Marks
• Provides a correct solution	2
• Shows probability that A wins on second turn is rq , or equivalent merit	1

Sample answer:

$$\begin{aligned}
 P(\text{A wins on 1st or 2nd}) &= p + rq \\
 &= p + r(1 - (p + r)) \\
 &= p + r - pr - r^2 \\
 &= p(1 - r) + r(1 - r) \\
 &= (1 - r)(p + r)
 \end{aligned}$$

Question 14 (b) (ii)

Criteria	Marks
• Provides a correct solution	3
• Observes limiting probability is a geometric series, or equivalent merit	2
• Finds probability that A wins on third turn, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 P(\text{A wins eventually}) &= p + rq + r^2p + r^3q + \dots \\
 &= p(1 + r^2 + r^4 + \dots) + rq(1 + r^2 + r^4 + \dots) \\
 &= (p + rq)(1 + r^2 + r^4 + \dots) \\
 &= (p + rq) \times \frac{1}{1 - r^2} \\
 &= \cancel{(1 - r)}(p + r) \times \frac{1}{(1 + r)\cancel{(1 - r)}} \quad (\text{using part(i)}) \\
 &= \frac{p + r}{1 + r}
 \end{aligned}$$

Mathematics Extension 1

2014 HSC Examination Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	2.8	PE3
2	1	5.7	PE2
3	1	17.3	HE3
4	1	6.6	PE6, PE2
5	1	16.3	PE3
6	1	15.5	HE4
7	1	14.4	HE3
8	1	18.1	PE3
9	1	16.2	PE3
10	1	4.3	PE2, P4

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	3	9.4	PE3, P4
11 (b)	2	18.2	HE3
11 (c)	2	15.3	HE4
11 (d)	3	11.5	HE6
11 (e)	3	1.4E	PE3
11 (f)	2	8.8, 12.5	P7, H3, PE5
12 (a) (i)	1	14.4	HE3
12 (a) (ii)	2	14.4	HE3
12 (b)	3	11.4, 13.6	H8
12 (c)	3	14.3E	HE5
12 (d)	2	17.3E	HE3, HE7
12 (e)	1	16.4	HE7, PE6
12 (f)	3	14.2E	HE3
13 (a)	3	7.4	HE2, PE6
13 (b) (i)	2	5.1, 8.8	P4, P7, PE6
13 (b) (ii)	1	14.1E	HE5, PE6
13 (c) (i)	2	6.7E, 9.6	PE3
13 (c) (ii)	1	9.6	PE3
13 (c) (iii)	3	9.6	PE3
13 (d) (i)	1	2.10	PE3, PE6
13 (d) (ii)	2	2.9, 2.10	PE3, PE6
14 (a) (i)	2	14.3E	HE3

Question	Marks	Content	Syllabus outcomes
14 (a) (ii)	3	14.3E	HE3
14 (a) (iii)	2	14.3, 5.7, 13.7	HE3, H5
14 (a) (iv)	3	5.2, 10.6	HE3, H5
14 (b) (i)	2	3.3	H5, HE7, PE6
14 (b) (ii)	3	3.3	H5, HE7, PE6